Note that we have not put any symbol corresponding to x^2. We further note that 4 changes of sign occurred (at x^6, x^4, x^1 and x^0).

Definition 3.2

A **change of sign in the coefficients is said to occur** at the jth power of x in a polynomial $P(x)$, if the coefficient of x^{j+1} and the coefficient of x^j (or) also coefficient of x^{j-1} coefficient of x^j are of different signs. (For zero coefficient we take the sign of the immediately preceding nonzero coefficient.)

From the number of sign changes, we get some information about the roots of the polynomial using **Descartes Rule**. As the proof is beyond the scope of the book, we state the theorem without proof.

Theorem 3.7 (Descartes Rule)

If p is the number of positive zeros of a polynomial $P(x)$ with real coefficients and s is the number of sign changes in coefficients of $P(x)$, then $s-p$ is a nonnegative even integer.

The theorem states that the number of positive roots of a polynomial $P(x)$ cannot be more than the number of sign changes in coefficients of $P(x)$. Further it says that the difference between the number of sign changes in coefficients of $P(x)$ and the number of positive roots of the polynomial $P(x)$ is even.

As a negative zero of $P(x)$ is a positive zero of $P(-x)$ we may use the theorem and conclude that **the number of negative zeros of the polynomial $P(x)$ cannot be more than the number of sign changes in coefficients of $P(-x)$ and the difference between the number of sign changes in coefficients of $P(-x)$ and the number of negative zeros of the polynomial $P(x)$ is even.**

As the multiplication of a polynomial by x^k, for some positive integer k, neither changes the number of positive zeros of the polynomial nor the number of sign changes in coefficients, we need not worry about the constant term of the polynomial. Some authors assume further that the constant term of the polynomial must be non zero.

We note that nothing is stated about 0 as a root, in Descartes rule. But from the very sight of the polynomial written in the customary form, one can say whether 0 is a root of the polynomial or not. Now let us verify Descartes rule by means of certain polynomials.

3.9.2 Attainment of bounds

3.9.2 (a) Bounds for the number of real roots

The polynomial $P(x)=(x+1)(x-1)(x-2)(x+i)(x-i)$ has the zeros $-1, 1, 2, -i, i$. The polynomial, in the customary form is x^5-2x^4-x+2. This polynomial $P(x)$ has 2 sign changes, namely at fourth and zeroth powers. Moreover,

$$P(-x)=-x^5-2x^4+x+2$$

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